Doyoon (Daniel) Kim

310-339-5193 | dkim927@uchicago.edu | linkedin.com/in/doyoondanielkim | github.com/danielk927

Education

University of Chicago — Chicago, IL
B.S. Computer Science — *Expected June 2028*

- Activities: Hong Kong National Kendo Team, Financial Markets Program, UCLA Math Directed Reading Program, Susquehanna Virtual Discovery Day
- Awards: LAHacks Best Hardware, UCLA Whopathon Best Overall

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. Computational Biology — *Transferred*

Experience

Undergraduate Research Assistant — January 2026 – Present
University of California, Los Angeles — *Los Angeles, CA*

- Engineered thermal performance curve models using rTPC (**R**) and Python (**Pandas** + **SciPy**) scripts, visualized results using **Matplotlib** to find thermal optimum zone in Drosophila; paper pending for *Science Advances*
- Created data preprocessing pipeline in **Pandas** and **NumPy** to clean and normalize heterogeneous sleep data

Engineering Intern — Oct 2025 – April 2026
Anto Biosciences (YC F25) — *San Francisco, CA*

- Developed **Python** ETL pipeline on MIT SuperCloud cluster for LLM-as-a-judge evaluation: implemented regex parsing, structured JSON extraction, rating normalization, and validation heuristics across 1,200+ papers
- Engineered system prompts with structured evaluation rubrics, boosted prediction accuracy from **46% to 73%**
- Authored 7 comprehensive data visualizations and scientific figures for preprints on tokenization methods in machine learning and deep reinforcement learning for data sparsification, papers pending for *NeurIPS* and *Nature*

Projects

Personal Website - Real-Time Multiplayer 3D World — September 2026 – Present

- Synced **16 concurrent players** per room in real time with zero client-server desyncs under load by building a server-authoritative **Node.js WebSocket** server with client-side prediction over a deterministic simulation
- Implemented **AWS** as infra in **CDK**, with WebSockets proxied via **CloudFront** to a self-healing **EC2** origin
- Ran the game DOOM in the browser by writing a **RISC-V** emulator and a **JIT compiler** to JavaScript that runs **18x faster** than its interpreter, and cross-compiling DOOM's **C** source bare-metal against a from-scratch libc

LLM Inference Engine - C++/CUDA Model Serving — September 2026 - Present

- Built a **C++/CUDA transformer inference runtime** with custom kernels, **cuBLAS** matrix multiplication, and KV-cached decoding; validated intermediate activations and output logits against a **PyTorch** reference
- Implemented **paged KV-cache allocation**, **continuous batching**, and **chunked prefill** for variable-length requests; built reproducible benchmarks measuring prefill latency, decode throughput, and GPU memory usage

Prior - Synthetic Data Engine for Biomedical Segmentation — August 2026 - Present

- Built a synthetic-data platform that converted measured cell morphology into **1,000+ labeled microscopy images**, reducing dataset preparation time by **39%** through procedural cell modeling and physics-based rendering
- Increased dataset generation throughput from **10 to 15 image–mask pairs/minute** by implementing parallel batch workers behind an asynchronous **FastAPI** service, with object storage and **PostgreSQL** metadata tracking

Sync - AI Neurorehabilitation Platform — May 2026 – August 2026

- Built 10+ gesture-controlled rehab games using **Three.js** and **MediaPipe** for real-time pose tracking, backed by **PostgreSQL** with row-level security for per-patient data isolation, deployed via **Vercel** + **Cloudflare**
- Deployed to 3 NGO pilots in Hong Kong with **1,400+** waitlist signups and received investment offers from 2 angels

Skills and Interests

Languages: Python, C++, C, JavaScript, TypeScript, HTML/CSS, R, SQL
Libraries/Frameworks: PyTorch, cuBLAS, CUDA, NumPy, Pandas, Matplotlib, MediaPipe, React, FastAPI, Node.js
Tools: Git, Docker, PostgreSQL, AWS (CDK, EC2, S3, CloudFront, IAM, CloudWatch)
Interests: Beli, Kendo, Cooking and Baking, Eating, Watching the NBA, Fashion